# Sumit Joshi

*Full-Stack Software Engineer*

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## Summary

Full-stack software engineer and B.Tech (AI & ML) student with hands-on experience shipping production-style web applications, real-time systems, and Retrieval-Augmented Generation pipelines. Freelanced across four client engagements building full-stack products and automating QA workflows that cut manual testing effort by 60%. Currently architecting Pacer, a codebase intelligence platform combining a graph + vector data model with an AI-powered Q&A bot. Published peer-reviewed research applying deep learning to agricultural pest detection.

## Education

**Graphic Era Hill University** — Dehradun, Uttarakhand
*B.Tech – Computer Science & Engineering (AI & ML), CGPA: 7.7/10* — *2023 – Present*

## Technical Skills

**Languages**: Python, C/C++, Java, JavaScript, TypeScript, SQL
**Frameworks & Libraries**: React, Next.js, Express.js, NestJS, FastAPI, Django, Node.js, Socket.io, TensorFlow, PyTorch, Scikit-learn
**Databases & Cloud**: PostgreSQL/pgvector, MongoDB, Redis, Pinecone, AWS (EC2)
**Developer Tools**: Git, GitHub, Docker, CI/CD, Linux, REST APIs, WebSockets, Postman, Playwright, Selenium
**Concepts**: Data Structures & Algorithms, System Design, OOP, RAG Pipelines, Multi-tenant RBAC, Agile/Scrum

## Experience

**Freelance Software Developer** — Dec 2023 – Mar 2024
*Full-Stack Developer & Automation Engineer* — *Dehradun, Uttarakhand*

- Automated end-to-end browser testing workflows using Playwright and Selenium, reducing manual QA effort by 60% across 3 client projects
- Developed RESTful web applications with FastAPI and React, integrating Redis caching that improved average API response times by 40%
- Grew from single-client projects to managing 4 client engagements end-to-end – requirements gathering, feature development, deployment, and post-launch support

## Projects

**Pacer – Codebase Intelligence Platform** | *Next.js, NestJS, Postgres/pgvector, BullMQ* | GitHub | Live — July 2026 – Present

- Architected a monorepo platform (Next.js, NestJS, BullMQ workers) ingesting GitHub webhooks and Slack events into a Postgres/pgvector store, modeling ownership, dependencies, and technical debt as a queryable graph; engineered ingestion workers to sustain 500+ webhooks/min
- Built a Retrieval-Augmented Generation (RAG) bot delivering cited answers to codebase questions plus an interactive dashboard (bus-factor graph, debt heatmap, blast-radius view); designed multi-tenant RBAC with row-level security and AES-256-encrypted OAuth tokens

**Maestro – Multiplayer Chess Platform** | *Next.js, TypeScript, PostgreSQL, Socket.io* | GitHub | Live — Feb 2025

- Architected a multiplayer chess platform using Next.js and Socket.io, building on prior real-time experience to enable game creation, matchmaking, and live board state synchronization
- Designed a PostgreSQL schema with indexed queries for game history and leaderboard rankings; created a web scraper aggregating chess resources from 7+ external sources

**Hellum – Autonomous Campus Tour Robot** | *Python, Raspberry Pi, Pinecone, Gemini API, RAG* | GitHub — Jan 2025

- Designed and assembled an autonomous robot on Raspberry Pi, programming sensor fusion with ultrasonic sensors and a camera module to navigate campus with 95% obstacle avoidance accuracy
- Implemented a Retrieval-Augmented Generation pipeline using Pinecone vector database and Gemini API, enabling the robot to answer location-specific queries in under 2 seconds

**Lavender – Real-Time Chat Application** | *MongoDB, Express.js, React, Node.js, Socket.io, JWT* | GitHub | Live — Oct 2023

- Developed a full-stack real-time messaging app on the MERN stack, implementing Socket.io for instant bidirectional communication – first hands-on project with WebSockets
- Secured user sessions with JWT-based authentication and persisted chat history in MongoDB, supporting 50+ concurrent connections with live notification delivery

## Certifications

Machine Learning Specialization – Stanford University / Coursera (8 weeks)
AWS Cloud Essentials – Amazon Web Services (3 weeks)
Programming with JavaScript – Meta / Coursera (4 weeks; December 24, 2023)
Robotics Specialization – Northwestern University (19 weeks)
Edge AI for Microcontrollers Specialization

## Achievements & Extracurricular

**Published Research Paper** (2025) – "V-IPDS: Vision-based Insect Predator Detection for Beehive Monitoring using YOLOv11n"
**IEEE Matrix Hackathon Finalist** (2025) – Led a 4-member team to the finals, presenting a working prototype to industry judges
**Graphethon 2.0 Finalist** (2024) – Advanced to finals in a national-level hackathon by GEU among 500+ participants
**NSS GEHU Volunteer** (2023 – Present) – Progressed from participant to coordinator, organizing outreach drives